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Infinite Sequences and Series

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Absolute Convergence and the Ratio and Root Tests

① Absolute Convergence

② Ratio and Root Tests

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- Given any series $\sum a_n$, we can consider the corresponding series

$$\sum |a_n| = |a_1| + |a_2| + |a_3| + \cdots$$

whose terms are the absolute values of the terms of the original series.

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Remark:

- Notice that if $\sum a_n$ is a series with positive terms, then $|a_n| = a_n$ and so absolute convergence is the same as convergence in this case.

Definition:

$\sum_{n=1}^{\infty} a_n$ is absolutely conv. if

$\sum_{n=1}^{\infty} |a_n|$ is conv.

Ex: $\sum_{n=1}^{\infty} \frac{\cos n}{n^2}$

$= \frac{\cos 1}{1} + \frac{\cos 2}{4} + \frac{\cos 3}{9} + \frac{\cos 4}{16}$

$= +$ $\quad - \quad - \quad -$

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so, I will check whether the series is absolutly conv.

i.e. I will test this series

$$\sum_{n=1}^{\infty} \left| \frac{\cos n}{n^2} \right|$$

$$-1 \leq \cos n \leq 1$$

$$\frac{|\cos n|}{|n^2|} \leq \frac{1}{|n^2|}$$

$$\sum_{n=1}^{\infty} \frac{|\cos n|}{|n^2|} \leq \sum_{n=1}^{\infty} \frac{1}{|n^2|} \quad , n \in [1, \infty)$$

$$\sum_{n=1}^{\infty} \left| \frac{\cos n}{n^2} \right| \leq \sum \frac{1}{n^2}$$

$\sum \frac{1}{n^2}$ p-series with $p=2$

so, it is conv.

so, by Limit comparsion test our series is conv.

$\Rightarrow \sum_{n=1}^{\infty} \frac{\cos n}{n^2}$ is absolutly convergent.

Theorem: If the series is absolutly conv. $\Rightarrow$ it is convergent.

$\sum a_n$ absolutly conv $\Rightarrow \sum a_n$ is conv.

Ex: $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n} = -1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \dots$

(this is an alternating series)

which is conv. as:

1. $b_{n+1} \leq b_n$

   $\frac{1}{n+1} \leq \frac{1}{n}$ ✓

2. $\lim_{n \to \infty} \frac{1}{n} = 0$ ✓

But the series of absolute value

$$\sum_{n=1}^{\infty} \left| (-1)^n \frac{1}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n}$$

p-series with $p=1$ (Harmonic series) which is div.

From this example we get that we might have a series which is convergent but it is not absolutly conv.

$$\sum a_n \text{ is conv} \not\Rightarrow \sum a_n \text{ is absolutely conv}$$

The Ratio test

(1) $$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = L$$

If $L < 1$ then the series is absolutly conv.
$\Rightarrow$ i.e $\sum a_n$ is conv

If $L = 1$ then the ratio test is inconclusive (no conclusion can be given From this test and you should use onther test

If $L > 1$ then the series $\sum a_n$ is div.

Example:

$$\sum \frac{3^n}{n!}$$

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Sol:

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty}$$

$$a_{n+1} = \frac{3^{n+1}}{(n+1)!} \quad , a_n = \frac{3^n}{n!}$$

$$= \lim_{n\to\infty} \left| \frac{\frac{3^{n+1}}{(n+1)!}}{\frac{3^n}{n!}} \right| =$$

$$\lim_{n\to\infty} \left| \frac{3^{n+1}}{(n+1)!} \cdot \frac{n!}{3^n} \right|$$

$$= \lim_{n\to\infty} \left| \frac{3^{n+1}}{3^n} \cdot \frac{n!}{(n+1)!} \right|$$

$$= \lim \left| \frac{3^n \cdot 3}{} \cdot \frac{n!}{} \right|$$

$$= \lim_{n\to\infty} \left| \frac{3^n}{(n+1)(n!)} \right|$$

$$= \lim_{n\to\infty} \left| \frac{3}{n+1} \right| = 3 \lim_{n\to\infty} \frac{1}{n+1} = 3\cdot(0) = 0$$

$$\Rightarrow \quad L = 0 < 1$$

By ratio test our series is absolutly conv. $\Rightarrow$ it is con.

Remark: This information For math III

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e$$

?

Example: Test the convergence

For the series

$$\sum_{n=1}^{\infty} \frac{n^n}{n!}$$

Sol:

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| =$$

$$= \lim_{n\to\infty} \left| \frac{\frac{(n+1)^{n+1}}{(n+1)!}}{\frac{n^n}{n!}} \right|$$

$$= \lim_{n\to\infty} \left| \frac{(n+1)^{n+1}}{(n+1)!} \cdot \frac{n!}{n^n} \right|$$

$$= \lim_{n\to\infty} \left| \frac{(n+1)^{n+1}}{(n+1)\,n!} \cdot \frac{n!}{n^n} \right|$$

$$= \lim \left| \frac{(n+1)^{n+1}}{} \cdot \frac{1}{} \right|$$

$$\lim_{n\to\infty} \left| \frac{n^n}{n+1} \right|$$

$$= \lim_{n\to\infty} \left| \frac{(n+1)(n+1)^n}{n^n} \cdot \frac{1}{(n+1)} \right|$$

$$= \lim_{n\to\infty} \left| \left(\frac{n+1}{n}\right)^n \right|$$

↳ just distribute the den. over the num's

$$= \lim_{n\to\infty} \left| \left(1 + \frac{1}{n}\right)^n \right|$$

$$= e = 2.7 = L > 1$$

$\Rightarrow$ By the Ratio test our series is divergent

Example 1

- Determine whether the following series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots$$

- is absolutely convergent or not

- is a convergent *p*-series ($p = 2$).

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Solution:

- as

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n^2} \right| = \sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots$$

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Example 2

- We know that the alternating harmonic series

- is convergent, but it is not absolutely convergent because the corresponding se $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$ nt

- which is the harmonic series (p-series with $p = 1$) and is

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2 Definition A series Σa_n is called **conditionally convergent** if it is convergent but not absolutely convergent.

- Example 2 shows that the alternating harmonic series is conditionally convergent. Thus it is possible for a series to be convergent but not absolutely convergent.

- However, the next theorem shows that absolute convergence implies convergence.

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Example 3

- Determine whether the series

- is convergent or divergent.

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Solution:

This series has both positive and negative terms, but it is not alternating. (The first term is positive, the next three are negative, and the following three are positive: The signs change irregularly.)

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Example 3 – Solution

cont'd

- We can apply the Comparison Test to the series of absolute values

- Since $|\cos n| \leq 1$ for all n, we have

- We know that $\sum 1/n^2$ is convergent (p-series with $p = 2$) and therefore $\sum |\cos n|/n^2$ is convergent by the Comparison Test.

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Example 3 – *Solution*

cont'd

- Thus the given series $\sum (\cos n)/n^2$ is absolutely convergent and therefore convergent by Theorem 3.

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Absolute Convergence and the Ratio and Root Tests

- The following test is very useful in determining whether a given series is absolutely convergent.

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Absolute Convergence and the Ratio and Root Tests

- **Note:**
  Part (iii) of the Ratio Test says that if the test gives no information. For instance, for the convergent series $\sum 1/n^2$ we have

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{\frac{1}{(n+1)^2}}{\frac{1}{n^2}} = \frac{n^2}{(n+1)^2} = \frac{1}{\left(1+\frac{1}{n}\right)^2} \to 1 \qquad \text{as } n \to \infty$$

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- whereas for the divergent series $\sum 1/n$ we have

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{\dfrac{1}{n+1}}{\dfrac{1}{n}} = \frac{n}{n+1} = \frac{1}{1+\dfrac{1}{n}} \to 1 \quad \text{as } n \to \infty$$

- Therefore, if the series $\sum a_n$ might converge or it might diverge. In this case the Ratio Test fails and we must use some other test.

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Example 5

- Test the convergence of the series

$$\sum_{n=1}^{\infty} \frac{n^n}{n!}.$$

- If you know that

$$\left(\frac{n+1}{n}\right)^n = \left(1+\frac{1}{n}\right)^n \to e$$

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Absolute Convergence and the Ratio and Root Tests

- The following test is convenient to apply when *n*th powers occur.

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